

A TRIPTYCH ARTICLE

Against the Instrumentalists

Verification, Its Limits, and the Reach of Reason

Contents

1	The Case for the Instruments	1
2	The Criterion Meets Itself	2
3	What the Laboratory Already Knows	3
4	From Effects to Their Ground	4
5	What the Church Stakes on Reason	5
6	The Sciences Given Back	6
	Notes	8
A	Scope, Method, and Qualifications	10
B	References	12
C	Generation Metadata	12

1 The Case for the Instruments

A cesium fountain clock will lose about one second in a hundred million years. Two laboratories on opposite sides of the world, staffed by people who share no language, politics, or religion, will read the same interferometer the same way, and when their numbers disagree past the stated uncertainty, both teams treat the disagreement as a fact to be explained rather than an opinion to be tolerated. There is nothing else in human life quite like this. Courts, parliaments, and philosophy departments should all envy it.

Set beside that record the history of the questions instruments cannot touch. Whether the world had a beginning, whether the soul survives the body, whether anything necessary stands beneath the contingent things we see—these questions are ancient, and the disputes about them are ancient too. No electrometer has ever moved when an argument for the existence of God succeeded. The debates that measurement entered, measurement largely settled; the debates it could not enter appear, from a certain distance, never to have moved at all.

From this asymmetry a conclusion has repeatedly been drawn, and it deserves to be stated at full strength, because this essay exists to argue with it. The conclusion is not that empirical method is excellent, which no one here disputes. It is that empirical method is *exhaustive*: that what cannot in principle be brought before observation is not merely unsettled but unknowable, or not merely unknowable but not even a genuine question. Knowledge, on this view, has exactly one instrument panel, and whatever registers on no instrument belongs to poetry, psychology, or noise.

The pedigree of this position is distinguished. David Hume closed his *Enquiry Concerning Human Understanding* with the most famous book-burning order in philosophy:

When we run over libraries, persuaded of these principles, what havoc must we make? If we take in our hand any volume; of divinity or school metaphysics, for instance; let us ask, *Does it contain any abstract reasoning concerning quantity or number?* No. *Does it contain any experimental reasoning concerning matter of fact and existence?* No. Commit it then to the flames: For it can contain nothing but sophistry and illusion.¹

Note the shape of the verdict. The offending volume is not to be read and refuted; it is to be classified and destroyed. Two admissible kinds of content are named—relations of quantity, matters of experimental fact—and whatever fits neither is condemned without further hearing.

Auguste Comte gave the same verdict a philosophy of history. Each branch of knowledge, he announced, “passes successively through three different theoretical conditions: the Theological, or fictitious; the Metaphysical, or abstract; and the Scientific, or positive.”² Theology and metaphysics are not rivals to be answered but stages to be outgrown, as a man outgrows his childhood; the positive stage renounces “the vain search after Absolute notions, the origin and destination of the universe, and the causes of phenomena” and confines itself to laws, “their invariable relations of succession and resemblance.” On this account no argument against metaphysics is required, any more than an argument is required against being young.

The twentieth century sharpened the verdict into a criterion. The logical positivists of the Vienna Circle, and their most effective English advocate, A. J. Ayer, saw that calling metaphysics *false* concedes too much, since a false statement is still a statement. The stronger move was to deny that metaphysical sentences say anything at all. In *Language, Truth and Logic*, published in 1936 when he was twenty-five, Ayer laid down the rule: “The criterion which we use to test the genuineness of apparent statements of fact is the criterion of verifiability. We say that a sentence is factually significant to any given person, if, and only if, he knows how to verify the proposition which it purports to express.”³ Sentences that fail the test are not mistaken; they are “pseudo-propositions.” And Ayer did not leave the application to the reader’s imagination: “no statement which refers to a ‘reality’ transcending the limits of all possible sense-experience can possibly have any literal significance,” so that “the labours of those who have striven to describe such a reality have all been devoted to the production of nonsense.”

The consequence for religion was drawn with the same tidiness. Ayer’s position was not atheism, which he considered as meaningless as theism, since the atheist denies the very sentence the theist affirms and a denial of nonsense is nonsense too. “To say that ‘God exists’ is to make a metaphysical utterance which cannot be either true or false.”⁴ The believer is not wrong about the world. He has failed, while appearing to speak, to say anything about the world at all.

This family of positions needs a name, and the available technical labels—positivism, verificationism, scientism—each carry historical baggage this essay does not want to adjudicate. I will call its adherents *instrumentalists*: those who

convert the instruments of empirical science, and the observational tests they serve, from an excellent means of knowing into the sole criterion of what can be known or what is real.⁵ The definition matters. An instrumentalist in this sense need not be a working scientist, and most working scientists are not instrumentalists in this sense; the position is a philosophical thesis about knowledge, not a laboratory practice.

One more thing should be conceded before the argument begins, because it is true and because the position's adherents have earned it. Instrumentalism is almost always adopted as a discipline of honesty. Its attraction is not contempt for meaning but hatred of wishful thinking: the suspicion that where no test is possible, assertion expands to fill the available space, and that humanity has spent millennia mistaking the echo of its own hopes for the voice of reality. That suspicion is not contemptible. It is, in fact, a moral posture—a commitment about how one *ought* to form beliefs—and the irony of that fact will matter later. But a moral posture wrapped around a philosophical thesis must still submit the thesis to examination. The thesis is that only the empirically testable can be known. The examination begins with an awkward question: how would one test it?

2 The Criterion Meets Itself

Take the criterion as Ayer stated it: a sentence is literally significant only if the proposition it expresses is analytic—true in virtue of the meanings of its terms, like “all bachelors are unmarried”—or empirically verifiable. Now ask what kind of sentence *that* is.

It is not a report of observation. No one has seen meaninglessness through an eyepiece, and no experiment has an outcome that would confirm or refute the claim that unverifiable sentences lack meaning. Nor is it an empirical generalization about how people use words, for if it were, it would be refuted daily: speakers of every language confidently use, understand, paraphrase, and translate sentences about God, the soul, and the ground of the world, and an empirical linguistics would have to record that such sentences function meaningfully in every speech community on earth. The criterion was never meant as a description of usage; it was meant as a verdict on usage.

Is it then analytic—true by definition? Ayer, pressed on the point, took this way out: the criterion, he wrote in 1946, is put forward as a definition, though not an arbitrary one.⁶ But a definition is a proposal, and a proposal can be declined. If “meaning” is defined so that only the empirically testable has it, then the metaphysician loses the word and keeps everything the word used to name: his sentences still assert, still contradict one another, still invite belief and denial; only now some other term must be found for what they have. A stipulation can redraw a map; it cannot execute the inhabitants. The criterion was advertised as a discovery about language—something we had all somehow failed to notice about our own speech—and under examination it shrank to a resolution about how its adherents proposed to talk. Resolutions carry no anathemas. Hume's bonfire needed a warrant, and a warrant, on the criterion's own terms, is exactly what the criterion could not be: not analytic without becoming optional, not empirical without becoming false, and if neither, then by its own test not even meaningful. It is difficult to name another doctrine in the history of philosophy that so precisely describes the conditions of its own destruction.

The self-application problem was noticed early, and it was not the only wound. The criterion also proved impossible to formulate. If “verifiable” means *conclusively* verifiable by finite observations, then every universal law of nature is meaningless, since no finite set of observations conclusively establishes a claim about all cases; the criterion built to protect science would have condemned it root and branch. So the requirement was weakened: a sentence is significant if observation is *relevant* to it, if some experiential statements can be derived from it with the help of auxiliary assumptions. But that let everything back in, for with suitably chosen auxiliaries, observational consequences can be extracted from any sentence whatever, including the metaphysical ones the criterion was designed to expel. Ayer repaired the formulation in 1946; the repair was broken in its turn; each subsequent tightening either strangled physics or readmitted metaphysics.⁷ By 1950 Carl Hempel, himself a logical empiricist and one of the movement's most rigorous minds, had surveyed every proposed version and delivered what amounted to an obituary: no criterion of the required sharpness could be drawn, and cognitive significance was at best a matter of degree. Ayer himself, reissuing his book in 1946, opened the new introduction by conceding that “the questions with which it deals are not in all respects so simple as it makes them appear.”⁸ Understatement is also a philosophical genre.

Two features of this collapse deserve emphasis, because they determine what may and may not be concluded from it.

First, the criterion was not refuted by theologians defending their territory. It was dismantled by logicians and philosophers of science, most of them unsympathetic to metaphysics, who simply could not make the rule sort sentences the way it was supposed to. The verification principle died of internal injuries, among friends. When the

Catholic Church much later described positivism's central claim as one that "critical epistemology has discredited," it was reporting the coroner's finding, not issuing it.⁹

Second—and this must be said as plainly against the essay's own interest as possible—the collapse of the criterion does not prove that any metaphysical sentence is true, or even that all metaphysical sentences are meaningful. A rule that fails to condemn a prisoner does not thereby acquit him. Some candidate metaphysics is surely confused; some sentences that look like assertions surely are not. What the collapse removes is something narrower and more important: the *blanket* verdict, the classification that spared the instrumentalist the labor of reading the volume before burning it. Hume's librarian asked two questions of the book's spine. After Hempel, the book has to be opened, and each argument in it met the way arguments are met—by counterargument. For half a century, in departments where the positivist creed was taught as hygiene, a generation learned that the question of God had not been answered but *dissolved*, and that the dissolution was a result of modern logic. It was not. The dissolution failed by the standards of modern logic, and the question stands where it stood, unread and unanswered, waiting.

That is a purely negative result, and if the matter ended there, this essay would be an exercise in demolition. It does not end there. The criterion's failure has a cause, and the cause is more instructive than the failure. The rule could not draw a boundary around knowledge at the edge of the empirically testable because knowledge never had that boundary—not even inside the laboratory the rule was built to flatter.

3 What the Laboratory Already Knows

Watch what a measurement actually involves, and count the things known in the course of it that no instrument reports.

Begin with the arithmetic. A physicist averages five readings; the average requires that five is what the readings number, that addition behaves tomorrow as it behaved today, and that division does not depend on the operator's mood. No experiment established these truths, and none could overturn them: an apparent laboratory refutation of arithmetic would be treated—rightly—as a miscount, a malfunction, or a joke. The same holds for logic. The whole evidential force of an experiment runs through inference: *if* the theory is true, *then* the meter reads thus; the meter does not read thus; therefore the theory is in trouble. That pattern—the logician's *modus tollens*—is not a finding of any science. It is what makes findings possible. A mind that had to verify logic empirically could never get an empirical verification started, since the bearing of evidence on hypothesis *is* a logical relation.¹⁰

Continue with the reach of the conclusions. Science does not want yesterday's meter readings; it wants laws—claims about all bodies, all charges, every case, observed and unobserved, past and future. The inference from *tested here* to *holds everywhere* is called induction, and its warrant is famously not itself an observation. Hume, the patron of the flames, was also the man who showed that no experience can certify that unexperienced cases resemble experienced ones, since any argument from past success of induction to its future success is induction again.¹¹ The working scientist is right to trust induction. But the trust is a piece of rational confidence that outruns every possible instrument reading, held by every experimentalist as a condition of experiment.

Continue with the sociology of evidence. Almost nothing any scientist knows was verified by that scientist. She knows the mass of the electron from tables, the reliability of the tables from the practice of the community, the results of the crucial experiments from reports written by the dead. Her own data from last year she knows by memory and by trusting records she has no independent means to check against the past itself. Testimony, memory, the existence of other minds competent to testify—this is the load-bearing structure of scientific knowledge, and none of it registers on any gauge. A consistent instrumentalist, permitted to believe only what he could bring before observation, would be reduced to the contents of his present visual field; he could not even believe in the previous page of his own lab notebook, since the past that wrote it is exactly the kind of thing no present observation can contain.

Continue, finally, with the norms. The entire enterprise runs on the conviction that data ought to be reported honestly, that a beautiful theory must yield to an ugly fact, that fabricating results is not merely imprudent but *wrong*. These are judgments about obligation, and obligation casts no shadow on a detector. On Ayer's official account they are not judgments at all but expressions of feeling, so that "falsifying data is wrong" says no more than a groan says. No scientific institution could survive a week on that reading of its own rules, and none has ever tried. The instrumentalist's laboratory is held together by moral knowledge his theory classifies as noise; his position is, in the exact sense, parasitic—alive only because the knowledge it denies keeps feeding it.

And at the center of all of it sits the thing the instruments cannot be: the one who reads them. A needle against a scale is not information; it is brass and ink until a mind takes it *as* standing for a quantity, under a theory, within a calibration, toward a conclusion. Every measurement terminates in an act of understanding and judgment that is not itself a measurement. This is not mysticism about consciousness; it is the plainest description of what happens when anyone reads a dial. The instrument extends the senses magnificently. It does not extend one inch beyond the senses toward the judgment that makes sense of them.

Now assemble the inventory. Inside the most disciplined empirical practice humanity has built, we find knowledge of mathematics and logic, which no observation establishes; rational warrant for induction, which outruns all observation; knowledge by testimony and memory of a past and of other minds that no observation contains; moral knowledge of the norms of inquiry; and the knower's own understanding, presupposed by every reading and reported by none. None of this is exotic. All of it is knowledge, if anything is—the instrumentalist himself stakes his practice on every item. The thesis that only the instrument-testable can be known is therefore not a modest generalization from science; it is a description that fits no science, no scientist, and no single hour of scientific work that has ever been performed.

What follows is not that anything goes. The right conclusion is more precise, and it is the hinge of this essay: knowledge is *proportioned to its objects*. Quantities are known by measuring because quantity is the sort of thing measurement fits. Theorems are known by proof, particulars of the past by testimony and record, obligations by practical reason, minds by their expression in speech and act—and in each case the method answers to the kind of thing being known, and is answerable to it, testable in its own register, correctable by its own failures. This plurality is the opposite of relativism: it is rigor, distributed. The demand that every object submit to the method proper to quantity is not rigor but a category mistake enforced with a confident face.

One question, then, can no longer be dismissed on procedural grounds. Sensible, measurable, contingent things exist; the sciences know them with growing precision. Whether such things point beyond themselves—whether the existence of the measurable has a ground that is not itself one more measurable thing—is a question about reality, askable in earnest, condemned by no coherent criterion. It must be judged the way every serious question is judged: by the arguments. To those arguments, and to the tradition that spent two thousand years refining them, the essay now turns.

4 From Effects to Their Ground

The word “metaphysics” now needs rescuing from its caricature, because the caricature is what the instrumentalist refuted. In the caricature, the metaphysician closes his eyes to experience, consults a private faculty, and returns with news about a hidden world—assertion without discipline, precisely the wishful thinking the positivist hygiene was designed to kill. Whatever deserves that description deserves its fate. But the central tradition of Western metaphysics, and certainly the Catholic philosophical tradition, never worked that way, and said so with a bluntness that ought to embarrass the caricature.

Thomas Aquinas states the tradition's starting point in words an empiricist could sign: “Our natural knowledge begins from sense. Hence our natural knowledge can go as far as it can be led by sensible things.”¹² No innate blueprint of God, no private channel: the mind's material is what the senses deliver—the same stones, stars, and meter readings the sciences study. The question is only how far sensible things can *lead*. And Aquinas's claim, argued rather than announced, is that they lead beyond themselves, because they present themselves as effects. His statement of the method is almost aggressively sober: “from every effect the existence of its proper cause can be demonstrated, so long as its effects are better known to us... Hence the existence of God, in so far as it is not self-evident to us, can be demonstrated from those of His effects which are known to us.”¹³ Not seen in a vision; not felt in the heart—*demonstrated*, the cold syllogistic word, from effects, the empiricist's own material.

What kind of reasoning is this? Not physics, and not in competition with physics. Physical explanation characteristically runs from state to state: the present configuration of a system is explained by an earlier configuration plus laws. Metaphysical inquiry asks a question that survives every such explanation, however complete: why is there a system in any configuration, and why do laws govern anything, rather than nothing at all? The contingent thing before me—this oak, this instrument, this galaxy—does not exist by necessity; it began, it depends, it could fail to be. Trace its dependence to prior things and the question does not shrink; it accumulates, because a chain of borrowers explains no loan. Either the regress of dependent things terminates in something that does not borrow its existence—something whose nature is not *to be conditioned* but *to be*—or the existence of the whole ensemble hangs on nothing. That,

compressed past all its needed refinements, is the shape of the classical argumentation: not a laboratory report, but not an emotion either. It is inference from the manifest character of sensible things—their dependence, their contingency, their borrowed actuality—to what their existence requires. Whether it succeeds is a question of philosophical argument, and the arguments have been attacked and defended at the highest level of technical seriousness for centuries. What the arguments are *not* is meaningless; that verdict required a criterion, and the criterion is dead.¹⁴

Notice next how little is claimed, because the modesty is structural, not tactical. From effects, Aquinas insists, the mind cannot be led to see *what God is*: “the sensible effects of God do not equal the power of God as their cause,” so that from them “the whole power of God cannot be known; nor therefore can His essence be seen.” What can be known is narrower and harder: “we can be led from them so far as to know of God ‘whether He exists,’ and to know of Him what must necessarily belong to Him, as the first cause of all things, exceeding all things caused by Him”—that all creatures stand to Him as caused to cause; that He is none of the things He causes; and that creatures fall short of Him not through any defect in Him but because He exceeds them all.¹⁵ This is a knowledge made mostly of negations and relations—a silhouette, not a portrait. The tradition that is accused of overreaching wrote the limits of its own reach into its founding texts.

It is worth pausing on how exactly this answers the instrumentalist’s deepest instinct. His fear, remember, was assertion without discipline. But here is a discipline: premises drawn from public, sensible reality; inference by explicit steps, each attackable; conclusions bounded with almost legal care, claiming existence and relation while renouncing inner description. One may dispute a premise—many have—and the dispute will be philosophy, conducted with reasons. What one may not do, after the collapse of the criterion, is refuse the tribunal: to classify the argument as noise because its conclusion is not the sort of thing a detector could register. Of course it is not. The argument’s entire point is that the detector, and every detectable thing, is an effect; a ground of all detectable things would be exactly what no detector could contain. To demand that the first cause show up as one more item inside the world’s inventory is not rigor about God; it is a misunderstanding of what was ever being claimed.

The senses, then, are not despised in this tradition; they are trusted further than the empiricist trusts them—trusted to bear the weight of an inference that runs through them to what they cannot picture. The measurable world is not the rival of metaphysical reason but its raw material and its springboard. That conviction did not originate with Aquinas, and it is far older than the medieval schools. Before it was a philosophy it was a scriptural claim—a claim, remarkably, about what human reason can do at its best and what it is culpable for failing to do. The essay has argued on philosophy’s ground to this point. It now reports, as the next witness, what the Catholic tradition has actually taught—because the teaching turns out to be the last thing an instrumentalist would predict a church to say.

5 What the Church Stakes on Reason

A reasonable person schooled on the instrumentalist picture expects religion to relate to reason roughly as a debtor relates to an auditor: evasively. Faith, on that picture, is what you assert when you cannot show; a church, of all institutions, should be the last to bind itself publicly to the competence of unaided human reason. It is therefore worth watching, with some care, what the Catholic Church has actually committed itself to—not in apologetic pamphlets but in Scripture read as her own, and in the most solemn register her teaching possesses.

The claim appears first inside the Bible itself, and it appears as a *rebuttal*. The Book of Wisdom, written within a century or so of Christ, surveys the nature-worship of the Gentile world: “all men are vain, in whom there is not the knowledge of God: and who by these good things that are seen, could not understand him that is.” The complaint is not that pagans reasoned about the world but that they stopped reasoning too soon, resting in fire, wind, stars, or sun: “For by the greatness of the beauty, and of the creature, the creator of them may be seen, so as to be known thereby.” And then the sentence that could serve as this essay’s epigraph, aimed at the natural philosophers of the author’s day: “For if they were able to know so much as to make a judgment of the world: how did they not more easily find out the Lord thereof?”¹⁶ The verse grants the scientists of antiquity their competence—they could gauge the world—and indicts them precisely for the inference they declined to complete. Ability to measure was never the issue. The issue was stopping at the measurable.

Paul, plainly aware of this tradition, universalizes it in the opening argument of Romans: “the invisible things of him, from the creation of the world, are clearly seen, being understood by the things that are made; his eternal power also, and divinity: so that they are inexcusable.”¹⁷ *Understood by the things that are made*: knowledge of the invisible, routed through the visible—the exact structure Aquinas would later defend with syllogisms, asserted here as the

reason idolatry is “inexcusable.” Scripture’s quarrel with paganism is not that it looked at nature too closely but that it did not look *through* nature far enough.

Eighteen centuries later, the First Vatican Council raised this scriptural conviction to the rank of solemn dogma—and the timing deserves attention. The year was 1870; Comte’s positivism was ascendant, and the intellectual pressure on Christianity was precisely the instrumentalist pressure this essay has examined. A church in retreat would have hedged. Instead, the council defined: “Eadem Sancta Mater Ecclesia tenet et docet, Deum, rerum omnium principium et finem, naturali humanae rationis lumine e rebus creatis certo cognosci posse”—in the rendering the Church’s own Catechism now gives, “God, the first principle and last end of all things, can be known with certainty from the created world by the natural light of human reason”—and the council cites Romans 1:20 as its warrant.¹⁸ And it attached a canon: whoever says that the one true God *cannot* be known with certainty by the natural light of reason from the things that are made stands condemned.

Stop and take the measure of that. The anathema does not fall on the atheist here; it falls on the *fideist*—the believer who, despairing of arguments, would rest faith on feeling and dismiss reason’s approach to God as impossible. The Catholic Church’s dogmatic response to the century of positivism was to bind her own children, under solemn censure, to the competence of natural reason. She placed herself, permanently and irrevocably, on reason’s side of the quarrel—against the instrumentalist who shrinks reason to the measurable, and equally against the pious skeptic who agrees with the instrumentalist about reason and merely adds a leap. It is difficult to imagine an institutional act less like the caricature of faith as the enemy of thought.

The definition is also carefully bounded, and the boundaries matter as much as the claim. It asserts a *capacity*—*cognosci posse*, can be known—not that any particular philosophical proof is dogma, not that every mind actually completes the ascent, not that natural knowledge suffices for salvation. The same constitution immediately adds that God chose also to reveal Himself, and the Catechism explains why both are needed: in the concrete conditions of human life, the mind is “hampered in the attaining of such truths, not only by the impact of the senses and the imagination, but also by disordered appetites”; man therefore “stands in need of being enlightened by God’s revelation” even about truths reason could in principle reach, so that they may be known “with ease, with firm certainty and with no admixture of error.”¹⁹ Revelation, in this account, is not a substitute for reason but a mercy to reason as it actually exists—burdened, hurried, and biased. Nor does the tradition fear a collision between the two: the same council teaches that “nulla tamen umquam inter fidem et rationem vera dissensio esse potest”—between faith and reason there can never be true disagreement—since the same God is the author of both.²⁰

Modern teaching has, if anything, sharpened the point against the position this essay has been examining. John Paul II’s encyclical on faith and reason names the adversary outright: scientism, “the philosophical notion which refuses to admit the validity of forms of knowledge other than those of the positive sciences,” which “relegates religious, theological, ethical and aesthetic knowledge to the realm of mere fantasy”—and observes, as this essay observed, that its positivist ancestor’s central claim is one that “critical epistemology has discredited.” What the same encyclical asks of philosophy is exactly what the argument of the preceding sections cleared room for: “a philosophy of genuinely metaphysical range, capable, that is, of transcending empirical data in order to attain something absolute, ultimate and foundational in its search for truth.”²¹

The convergence should be stated with epistemic care, because this essay has promised such care throughout. Sections above argued, on purely philosophical ground, that the exclusivist criterion destroys itself and that reason’s reach extends beyond the measurable; nothing in those arguments borrowed authority from Scripture or councils. What this section adds is a fact about the Catholic tradition: that it teaches, at the level of solemn definition, what the philosophical argument independently concluded—that reason, beginning from created things, can reach their ground. The agreement of the two lines is not offered as proof of either. But a reader who has been told all his life that faith and reason pull in opposite directions is entitled to notice where the parties actually stand. On the capacity of reason, the Church turns out to be the maximalist, and the instrumentalist the man of little faith.

6 The Sciences Given Back

Nothing in this essay has been an argument against science, and it is worth ending by saying exactly why—because the instrumentalist’s most durable rhetorical asset is the suggestion that to question his philosophy is to menace the laboratory.

The suggestion has things backwards. The empirical sciences do not assert instrumentalism; they do not need it; they were built, most of them, by people who would have found it unintelligible. What the sciences require is much more modest and much more secure: that nature is intelligible, that observation disciplines theory, that quantity yields to measurement, that honest method converges. Every one of those commitments survives this essay untouched—several of them, as the third section argued, turn out to be *stronger* than instrumentalism can account for, since they rest on logical, inductive, and moral knowledge that no instrument delivers. Relieving science of the exclusivist philosophy bolted onto it removes no result, retracts no method, and dims no achievement. It removes only a claim that was never a scientific claim at all: the philosophical thesis that the method's writ runs to the whole of reality. Chemistry loses nothing when it stops being asked to serve as a theory of meaning.

And the Catholic tradition, whatever its sins in particular episodes, is on record—in its most binding voice—as wanting science whole rather than diminished. A church that solemnly defines the competence of natural reason has no standing to fear reason's disciplined exercise on nature, and the definition of 1870 forecloses the fideism that would try. The believer who sneers at measurement betrays *Dei Filius* as surely as the positivist who worships it; both have accepted the same false premise, that the measurable and the real are rivals for the same throne. They are not rivals. On the tradition's account the measurable world is creation—the very thing through which, Wisdom and Paul and Aquinas and the council all insist, the invisible God is reached. Every improvement in our knowledge of the effects is, in principle, an enrichment of the premise from which the ascent begins. That is why the sciences sit more securely inside this tradition than inside instrumentalism, which can honor them only by first misdescribing them.

What, then, is the reader entitled to take away? Not that God's existence has been demonstrated in these pages; it has not been attempted, and honesty about that boundary has been a discipline of the whole essay. The demonstrations exist, they are serious, and they must be weighed as arguments—which is precisely the labor the instrumentalist criterion promised to spare us and could not. Nor is the takeaway that every metaphysical sentence deserves respect; the collapse of a bad filter is not the vindication of everything it caught. The takeaway is a restored question. The claim that only the measurable is real was itself unmeasurable; the claim that only verifiable statements are meaningful was itself unverifiable; the creed of the instruments could not survive being pointed at itself. What stands after its collapse is what stood before it: a contingent world that does not explain itself, a mind whose reach demonstrably exceeds its gauges, and the oldest question in philosophy, no longer dismissible by rule—asked now, as it was asked of the ancient star-gazers, with the same quiet pressure: *if you were able to know so much as to make a judgment of the world, how did you not more easily find out the Lord thereof?*

The instruments are good. They were never the judges. Read the dial, honor the reading, and then—because you are the kind of being that can—ask what the existence of dials, readings, and readers requires. That question is not a failure of intellectual nerve. It is intellectual nerve, resumed.

Notes

¹David Hume, *An Enquiry Concerning Human Understanding*, sect. XII, part III, final paragraph (E 12.34; SBN 165). Wording checked 2026-07-24 against the critical text at davidhume.org. The *Enquiry* was first published in 1748; the text is in the public domain.

²Auguste Comte, *Cours de philosophie positive* (1830–1842), opening chapter, as condensed and translated by Harriet Martineau, *The Positive Philosophy of Auguste Comte* (London: George Bell & Sons, 1896), vol. 1, pp. 1–2. Wording checked 2026-07-24 in a scan of the 1896 Bell printing. Martineau’s condensation, which Comte himself endorsed, is in the public domain.

³A. J. Ayer, *Language, Truth and Logic* (London: Victor Gollancz, 1936), ch. 1, “The Elimination of Metaphysics.” Wording checked 2026-07-24 in a scan of the Pelican reprint. Sentences with no empirical cash value may retain *emotional* significance, Ayer allows, but not *literal* significance. Brief quotations from this copyrighted work appear here under ordinary fair-quotation practice.

⁴Ayer, *Language, Truth and Logic*, ch. 6, “Critique of Ethics and Theology.” Ayer is explicit that his view is neither atheism nor agnosticism: the agnostic treats “God exists” as a genuine but undecidable proposition, whereas Ayer denies it is a proposition at all. The same chapter dissolves moral judgments into expressions of feeling—a consequence worth noticing, since it shows how much besides theology the criterion condemns.

⁵The term is coined for this essay and is defined only as stated. It is *not* the doctrine called instrumentalism in the philosophy of science—the view, associated with Duhem and others, that scientific theories are calculating devices for organizing observations rather than descriptions of unobservable reality. That doctrine concerns the interpretation of science and is untouched by this essay. Nor is the target empiricism as such, the thesis that our natural knowledge originates in experience (a thesis this essay will defend in its Thomistic form), nor the methodological naturalism by which the empirical sciences rightly confine themselves to natural causes within their own work. The target is the *exclusivist* conversion of empirical method into the universal test of meaning, knowledge, or reality.

⁶Ayer, *Language, Truth and Logic*, introduction to the second edition (1946; printed as an appendix in the Pelican issue), opening pages of the section “The Principle of Verification.” Checked 2026-07-24 in the same scan. Ayer proposes that the principle be treated as a definition rather than an empirical hypothesis about meaning, while claiming it is faithful to what “meaning” ordinarily means in scientific and everyday factual discourse—a claim the ordinary use of religious and metaphysical language sits awkwardly beside.

⁷The wounds were inflicted mostly from inside the empiricist camp: Isaiah Berlin, “Verification” (1939), and Alonzo Church’s 1949 review of Ayer’s second edition showed that both the original weak criterion and its repaired successor admitted arbitrary sentences. Carl G. Hempel’s survey, “Problems and Changes in the Empiricist Criterion of Meaning,” *Revue Internationale de Philosophie* 4, no. 11 (1950): 41–63 (reprinted in Leonard Linsky, ed., *Semantics and the Philosophy of Language* [Urbana: University of Illinois Press, 1952], 163–185, and revised in Hempel’s *Aspects of Scientific Explanation* [New York: Free Press, 1965]), walked through the successive formulations—complete verifiability, complete falsifiability, translatability into an empiricist language—and found each one either excluding manifest science (universal laws, mixed quantifications, theoretical terms) or excluding nothing. Hempel concluded that cognitive significance admits of degrees and cannot be captured by any sharp verifiability or falsifiability test. This narrative of the criterion’s career was checked 2026-07-24 against the Stanford Encyclopedia of Philosophy’s article “Carl Hempel”; Hempel’s 1950 text itself was identified but not independently collated for this essay, and no wording from it is quoted.

⁸Ayer, introduction to the second edition, first paragraph. In later decades Ayer’s assessments of the book grew harsher still; those remarks circulate in interview form and are deliberately not relied on here. The 1946 concession, printed in his own book, is sufficient for the argument.

⁹John Paul II, encyclical *Fides et ratio* (14 September 1998), no. 88: scientism “is the philosophical notion which refuses to admit the validity of forms of knowledge other than those of the positive sciences,” relegating “religious, theological, ethical and aesthetic knowledge to the realm of mere fantasy”; “In the past, the same idea emerged in positivism and neo-positivism, which considered metaphysical statements to be meaningless. Critical epistemology has discredited such a claim...” Official English text at vatican.va, checked 2026-07-24. The encyclical is cited here only as a witness that the Church’s response engages this exact position; its doctrinal use comes later in the essay.

¹⁰An empiricist of Quine’s stripe replies that logic and mathematics are not known apart from experience but are simply the most protected members of a single web of belief that faces experience as a whole, revisable in principle like everything else (W. V. Quine, “Two Dogmas of Empiricism,” *Philosophical Review* 60 [1951]: 20–43; cited from the standard bibliographic record, not collated for this essay). The reply is serious, but notice what it concedes and what it costs: it concedes that nothing is verifiable sentence-by-sentence, which was the criterion’s whole mechanism; and any deliberate *revision* of the web must itself be conducted by inference—some norm of consistency decides what “facing experience” requires—so the authority of logic reappears inside the very act said to revise it. Either way, the instrumentalist’s sharp test is gone.

¹¹Hume, *Enquiry*, sect. IV. The point here is not Humean skepticism—this essay takes induction to be rationally warranted—but the location of the warrant: whatever justifies trusting nature’s uniformity, it is not delivered by an instrument, since instruments report actual cases only and the uniformity claim outruns every actual case.

¹²Thomas Aquinas, *Summa theologiae* I, q. 12, a. 12, corpus: “naturalis nostra cognitio a sensu principium sumit, unde tantum se nostra naturalis cognitio extendere potest, inquantum manuduci potest per sensibilia.” Latin checked 2026-07-24 at the Corpus Thomisticum online corpus; English from the public-domain translation of the Fathers of the English Dominican Province, checked the same day at New Advent. Work, question, and article govern over either website’s presentation.

¹³*Summa theologiae* I, q. 2, a. 2, corpus. Aquinas distinguishes demonstration *propter quid* (through the cause, explaining why) from demonstration *quia* (from the effect back to the cause’s existence). Only the second is claimed for natural knowledge of God. Latin checked at Corpus Thomisticum, English at New Advent, 2026-07-24, as above.

¹⁴This essay defends the admissibility and rational character of such reasoning; it deliberately does not execute any demonstration or claim that the reader is now in possession of one. The five ways of *Summa theologiae* I, q. 2, a. 3, and their modern reconstructions and criticisms, are a literature of their own and lie outside this essay’s scope. Nothing here argues from gaps in scientific explanation; the reasoning at issue begins from the existence and dependence of things, not from phenomena science has not yet explained—a distinction developed in the Scope appendix.

¹⁵*Summa theologiae* I, q. 12, a. 12, corpus, in the English Dominican translation as above. The doctrine of analogical rather than univocal naming of God, which disciplines all such language, is at I, q. 13, and is noted here without development.

¹⁶Wisdom 13:1, 5, 9, Douay–Rheims translation (Challoner revision; public domain), checked 2026-07-24 at drbo.org. Wisdom belongs to the Catholic canon; readers of traditions that place it among the apocrypha may treat it here as an important pre-Christian Jewish witness. Verses 6–8 balance the rebuke with a striking gentleness: those who erred while “seeking God, and desirous to find him” are “less to be blamed”—the fault is resting in the search’s instruments rather than following them through.

¹⁷Romans 1:20; cf. 1:19, Douay–Rheims, checked 2026-07-24 at drbo.org. Exegetical honesty requires noting Paul’s purpose: the passage grounds a charge of culpability—the suppression of an available knowledge—within an indictment of idolatry; it is not a treatise on philosophical method. The reading of the verse as warrant for a real natural knowability of God from creatures is the Church’s received reading, made explicit when *Dei Filius* cites this very text for its definition and when the Catechism (CCC 32) appeals to it.

¹⁸First Vatican Council, dogmatic constitution *Dei Filius* (24 April 1870), ch. 2, *De revelatione*, with the corresponding canon: “Si quis dixerit, Deum unum et verum, Creatorem et Dominum nostrum, per ea, quae facta sunt, naturali rationis humanae lumine certo cognosci non posse; anathema sit” (canons on revelation, no. 1)—in working English, “If anyone says that the one true God, our Creator and Lord, cannot be known with certainty by the natural light of human reason through the things that are made, let him be anathema” (gloss by this essay; the Latin governs). Latin checked 2026-07-24 at the Holy See’s Latin text at vatican.va; the English rendering of the chapter 2 definition is that of the official English Catechism, no. 36, checked the same day.

¹⁹CCC 37–38, quoting Pius XII, *Humani generis*; official English at vatican.va, checked 2026-07-24. CCC 31 describes the classical arguments as “converging and convincing arguments” rather than “proofs in the sense of proofs in the natural sciences”; CCC 32–34 sketch the two starting points, the world (movement, becoming, contingency, order, beauty) and the human person (openness to truth, moral conscience, longing for the infinite); CCC 35 and 39 draw the consequence that reason’s capacity grounds both the credibility of faith and the Church’s ability to speak with unbelievers. Checked at the same source and date.

²⁰*Dei Filius*, ch. 4 (working gloss; Latin checked as above). The chapter locates apparent conflicts in either misstated dogma or errors paraded as conclusions of reason—a diagnosis that cuts, on even terms, against both bad theology and bad philosophy.

²¹*Fides et ratio*, nos. 88 and 83, official English at vatican.va, checked 2026-07-24. No. 88 is authoritative papal teaching engaging a philosophical position; its philosophical claims persuade by the arguments that support them, which this essay has tried to supply independently.

A Scope, Method, and Qualifications

A.1 Reader, question, and method

This is a discursive philosophical and theological article for a serious general reader who respects the empirical sciences and may hold, explicitly or by unexamined habit, that empirical testability is the boundary of the knowable or the real. Its governing question is whether that exclusivist thesis—named “instrumentalism” in these pages by a deliberate coinage—survives examination, and what standing metaphysical reasoning about God retains once it does not. The article’s claim is that the thesis fails by self-application and by misdescribing the sciences it honors; that knowledge is proportioned to its objects; that reasoning from sensible effects toward their ground is therefore admissible and disciplined; and that the Catholic Church is dogmatically committed to the competence of natural reason so to reason. The article clears and defends this space; it does not execute a demonstration of God’s existence.

The argument proceeds in epistemic order: the instrumentalist case at full strength; its internal collapse on philosophical grounds alone; the epistemology of scientific practice; the Thomistic account of knowledge from sensible effects; and only then the received scriptural and dogmatic witnesses, cited as the Church’s teaching rather than as conclusions of the preceding philosophy. English is the working language; no liturgical rite, civil or canonical jurisdiction, or mutable discipline governs any conclusion. Online witnesses were checked on 2026-07-24.

Source class	Function in the article	Governing boundary
Positivist and empiricist primary texts	Hume, Comte, and Ayer supply the exclusivist position in its classic formulations, quoted briefly and at exact loci.	The texts are engaged as arguments, not psychoanalyzed; the coined target is defined independently of any one author’s full position.
Philosophy-of-science scholarship	Hempel’s survey and the standard critical literature establish the historical collapse of the verifiability criterion.	Hempel’s 1950 paper was identified but not collated; the collapse narrative is checked against a named encyclopedic witness and Ayer’s own 1946 introduction.
Thomistic philosophy	ST I, q. 2, a. 2 and q. 12, a. 12 govern the adopted account of demonstration from effects and the sense-origin and limits of natural knowledge.	Adopted as the article’s philosophical framework, not asserted as dogma; the five ways themselves are outside scope.
Sacred Scripture	Wisdom 13 and Romans 1:19–20 witness the biblical claim that God is knowable from created things.	Quoted in the Douay–Rheims translation; exegetical limits of Romans 1 are stated beside the claim.
Authoritative Catholic teaching	<i>Dei Filius</i> ch. 2 and its first canon on revelation, CCC 31–39, and <i>Fides et ratio</i> 83 and 88 govern the doctrinal frame.	Dogmatic force is confined to what the definition states; no particular proof is presented as dogma.
Project synthesis	The coined term, the presupposition inventory, examples, orderings, and final formulations are original editorial synthesis.	Synthesis is argued from the checked sources and creates no doctrine; it is attributed to no source.

A.2 Included and excluded scope

Included are: the strongest form of the exclusivist appeal to scientific success; the self-application problem of the verifiability criterion and the criterion’s documented technical collapse; the knowledge presupposed by empirical practice (logic and mathematics, induction, testimony and memory, norms of inquiry, the judging knower); the proportioning of methods to objects; the Thomistic account of demonstration *quia* from sensible effects and its structural limits; the scriptural witness of Wisdom 13 and Romans 1; the dogmatic definition and canon of *Dei Filius* on natural knowledge of God, with CCC 31–39 and *Fides et ratio* 83 and 88; and the restoration of empirical method to its proper scope.

Excluded are: execution or evaluation of any demonstration of God’s existence, including the five ways; a history of logical positivism or of the Vienna Circle’s members; adjudication of realism versus instrumentalism in the philosophy of science; philosophy of mind and the hard problem of consciousness; foundations of mathematics; probabilistic, fine-tuning, or miracle-based apologetics; arguments from present gaps in scientific explanation; biblical criticism of the Book of Wisdom beyond its canonical status; and any claim about the salvation, culpability, or good faith of particular persons.

A.3 Material qualifications

1. “Instrumentalist” is a coined, reader-facing term defined in the first section. It is not the philosophy-of-science doctrine of instrumentalism, not empiricism as such, and not methodological naturalism; scientists as such are not its referent.
2. The collapse of the verifiability criterion does not establish the truth or meaningfulness of any metaphysical claim. The article draws only the negative conclusion that the blanket dismissal fails.
3. The self-application objection is presented with Ayer’s “definition” rejoinder, and the reply is answered on its merits; the objection is not treated as unanswerable by mere assertion.
4. The Quinean holist alternative to sentence-level verificationism is acknowledged and answered briefly; a full treatment of confirmation holism is outside scope.
5. The presupposition inventory claims that the listed items are knowledge not delivered by instruments; it does not claim they are unrevisable, innate, or independent of experience in every sense. The Thomistic sense-origin thesis is affirmed alongside it.
6. The contingency reasoning sketched in the fourth section is a compressed *shape* of classical argumentation, labeled as such; its needed refinements (against, for example, the fallacy-of-composition and brute-fact objections) are not supplied here.
7. Aquinas’s account is adopted as the article’s philosophical framework where Catholic theology permits schools; it is not presented as the only orthodox account of natural knowledge.
8. Romans 1:19–20 functions in Paul’s argument as grounding culpability for idolatry; the article’s use of it for natural knowability follows the Church’s received reading, explicitly so labeled.
9. *Dei Filius* defines the *capacity* of natural reason (*cognosci posse*); it does not canonize any particular proof, assert that all persons achieve such knowledge, or make natural knowledge sufficient for salvation. The article states these limits where the definition is quoted.
10. The canon’s English is a working gloss governed by the quoted Latin; the chapter’s English follows CCC 36’s official rendering.
11. *Fides et ratio* is authoritative papal teaching whose philosophical assessments bind as such teaching binds; the article supplies independent philosophical argument rather than resting the case on the encyclical’s authority.
12. The claim that the criterion “died among friends” concerns the documented technical literature; it does not assert that no religious thinker criticized positivism.
13. No claim is made that the Catholic Church’s historical record on particular scientific episodes is beyond criticism; the argument concerns her doctrinal commitments, not her administrative history.

A.4 Notes, translations, rights, currentness, and review

The numbered Notes carry exact citations, translation and edition identifications, verification dates, and qualifications that would interrupt the argument; they contain no missing premise. Scripture is quoted from the public-domain Douay–Rheims (Challoner) translation. *Dei Filius* is quoted from the Holy See’s Latin text with the Catechism’s official English for the chapter 2 definition and a labeled working gloss for the canon. Catechism and *Fides et ratio* quotations follow the official Vatican English web texts. Hume, Comte–Martineau, and the English Dominican translation of Aquinas are public domain; brief quotations from Ayer’s copyrighted *Language, Truth and Logic* appear under ordinary fair-quotation practice. Scripture, official texts, and third-party translations remain outside the project’s CC BY 4.0 grant; project-created prose, the coined terminology, and the organization are project content.

All online witnesses were checked on 2026-07-24; no mutable law or discipline is relied on, so no as-of fragility attaches to the conclusions. This revision received internal argumentative, source-consistency, quotation, rights, and production review by the authoring agent. Independent philosophical, theological, and ecclesiastical review is outstanding; no imprimatur, nihil obstat, or ecclesiastical approval is claimed, and internal checking is not independent review.

B References

Sacred Scripture

- Wisdom 13:1–9 and Romans 1:19–20, quoted from the Douay–Rheims translation (Challoner revision; public domain); working web witness checked 2026-07-24 at drbo.org (Wisdom 13) and drbo.org (Romans 1).

Authoritative Catholic witnesses

- First Vatican Council, dogmatic constitution *Dei Filius* (24 April 1870), ch. 2 *De revelatione* with canon 1 on revelation, and ch. 4 on faith and reason; Holy See Latin text, vatican.va, checked 2026-07-24.
- *Catechism of the Catholic Church* 31–35 (ways of coming to know God; “converging and convincing arguments”), official English; 36–38 (the knowledge of God according to the Church, quoting *Dei Filius* and *Humani generis*), official English; 39 (speaking of God with all men), official English. All checked 2026-07-24.
- John Paul II, encyclical *Fides et ratio* (14 September 1998), nos. 83 (philosophy of genuinely metaphysical range) and 88 (scientism); official English, checked 2026-07-24.

Thomistic framework

- Thomas Aquinas, *Summa theologiae* I, q. 2, a. 2 (demonstration *quia* of God’s existence from effects) and I, q. 12, a. 12 (sense-origin, reach, and limits of natural knowledge of God). Latin checked 2026-07-24 in the Corpus Thomisticum online corpus; English from the public-domain translation of the Fathers of the English Dominican Province via New Advent (q. 2) and New Advent (q. 12). Work, question, article, and division govern over either site’s presentation.

Positivist and empiricist primary texts

- David Hume, *An Enquiry Concerning Human Understanding* (1748), sect. IV (induction) and sect. XII, part III (the closing paragraph, E 12.34; SBN 165); public domain; critical web text checked 2026-07-24 at davidhume.org.
- Auguste Comte, *Cours de philosophie positive* (1830–1842), as condensed and translated by Harriet Martineau, *The Positive Philosophy of Auguste Comte*, vol. 1 (London: George Bell & Sons, 1896), pp. 1–3 (the law of the three stages); public domain; scan of the 1896 printing checked 2026-07-24 via the Internet Archive.
- A. J. Ayer, *Language, Truth and Logic* (London: Victor Gollancz, 1936; 2nd ed. 1946), ch. 1 (the verifiability criterion), ch. 6 (theology), and the 1946 introduction; in copyright, quoted briefly; wording checked 2026-07-24 in an Internet Archive scan of the Pelican reprint, which prints the 1946 introduction as an appendix.

Philosophy-of-science scholarship

- Carl G. Hempel, “Problems and Changes in the Empiricist Criterion of Meaning,” *Revue Internationale de Philosophie* 4, no. 11 (1950): 41–63; reprinted in Leonard Linsky, ed., *Semantics and the Philosophy of Language* (Urbana: University of Illinois Press, 1952), 163–185; revised as “Empiricist Criteria of Cognitive Significance: Problems and Changes” in *Aspects of Scientific Explanation* (New York: Free Press, 1965). Identified but not collated; its argument is used as reported in the following witness.
- James Fetzer, “Carl Hempel,” *Stanford Encyclopedia of Philosophy*, plato.stanford.edu, checked 2026-07-24, for the standard account of the criterion’s successive formulations and failure.
- Isaiah Berlin, “Verification,” *Proceedings of the Aristotelian Society* 39 (1938–39): 225–248, and Alonzo Church, review of Ayer’s *Language, Truth and Logic*, 2nd ed., *Journal of Symbolic Logic* 14 (1949): 52–53; cited from standard bibliographic records as the classic technical objections; not collated.
- W. V. Quine, “Two Dogmas of Empiricism,” *Philosophical Review* 60 (1951): 20–43; cited from the standard bibliographic record for the holist alternative; not collated.

C Generation Metadata

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